The Design and Implementation of a Sensorless Delta-Connected Brushless Direct Current Motor Drive for Home Appliances

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***Abstract*—The sensorless motor drive of Brushless Direct Current (BLDC) motors with wye-connected windings, now becomes accepted in many practical applications because it can solve the drawbacks of Hall sensors based motor drive. One of the most popular sensorless control schemes is based on the phase Back Electromotive Force (BEMF) or terminal phase voltage from the floated phase. Sensorless commutation signals are obtained from the detection of zero crossing points of the phase BEMF. However, it is impossible to adopt this technique for the delta-connected BLDC motors. Therefore, a sensorless motor drive with sensorless circuit based on the line-to-line terminal voltages for a delta-connected BLDC motor is proposed in this paper. Moreover, a low-pass filter with variable cut-off frequencies was designed successfully with a simple algorithm in this paper. Furthermore, the performance of the proposed sensorless control scheme was also compared to that of the Hall based control. Finally, the experimental results demonstrated that the sensorless motor drive was successfully developed for the delta-connected BLDC motor. The feasibility of the complete system was also validated.**

Keywords-brushless DC motor; sensorless motor drive; speed control; delta-connected winding

I. INTRODUCTION

In recent years, BLDC motors have been employed in many home appliances and industrial applications due to its excellent features such as high efficiency, long-life operation, and compactness [1]–[3]. In order to run a BLDC motor, position sensors are required to identify rotor position information for inverter switching control, and Hall-effect position sensors are typically used. However, it is well known that the Hall sensors are sensitive to temperature variations and harsh environments. This may lead to performance degradation. Moreover, the presence of Hall sensors might occupy a significant volume in certain applications in which its volume is limited. The installation of these sensors needs to be performed with high-precision [4]. These disadvantages can be solved with sensorless control methods. The sensorless control of BLDC motors have been deep investigated by many works [4]–[6]. Among these sensorless methods, one of the most popular techniques is the detection of Zero Crossing Points (ZCPs) of the BEMF from the floated phase. Accordingly, the wye-connected BLDC motors has been widely interested with a six-step switching interval technique. However, this approach cannot be used for BLDC motors with delta-connected winding because there is no free terminal phase voltage for capturing the BEMF with the delta-connected configuration. This configuration has not been adequately investigated in the previous studies for the sensorless control scheme. Therefore, a sensorless motor drive for the delta-connected BLDC motor was designed and implemented in this paper. The sensorless technique used was based on a detection circuit of the ZCPs of the average Line-to-Line (LL) terminal voltage instead of from the phase BEMF or phase terminal voltage. In [5], the sensorless control method was based on the mathematical model in which the LL voltage was estimated. The ZCPs detection algorithm was based on G function, and solved by programming. While in [7], [8], the Sensorless Commutation Signals (SCSs) were obtained from the ZCPs of the average LL voltage by using analog Low-Pass Filters (LPFs) combined with comparators to generate the proper SCSs. Accordingly, these LL ZCPs lag 30 electrical degree compared to ZCPs of the phase BEMF. It means that the LL ZCPs are exactly commutation points for current regulation of the inverter. This implies that this technique can be applied to the delta-connected BLDC motors as above mentioned. Regarding the sensorless control scheme with delta-connected winding, a digital phase-locked loop algorithm was proposed in [9]. It is based on the integration of the resultant BEMF, which is derived from the mathematical model of the delta-connected BLDC motor. This approach, however, is out of scope in this paper. Accordingly, in this paper, the sensorless circuit was based on the average LL terminal voltage technique with Schmitt trigger circuit [8] was

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adopted for the delta-connected BLDC motor with sinusoidal BEMF. However, of the sensorless circuit, the phase delay phenomenon caused by LPFs is inevitable. The cut-off frequency of the LPFs, in general, is designed for certain operating speeds. The phase delay angle will be reduced at high-speed if the value of the cut-off frequency is large. In contrast, if the cut-off frequency value is small, then the phase delay will be increased as increasing in speed. At low speeds, if the cut-off frequency value is high, subsequently the quality of SCS is poor caused by noise. This leads to decreasing in the performance of the motor or even starting failure caused by commutation error. This implies that it is difficult to design an analog LPF to operate with a fixed cut-off frequency for a wide speed-range. Consequently, in this paper, the LPFs was designed to operate with changeable cut-off frequencies for a wide speed-range.

Accordingly, the aim of this paper is to design the speed control of a sensorless delta-connected BLDC motor drive for home appliances such as fan, range-hood, blower, etc. Therefore, the rest of the paper is organized as follows. The mathematical model of the BLDC motor is presented in section two. The sensorless control scheme is introduced in section three. Section four is the system architecture. Experimental results and discussions are shown in section five. Finally, the conclusion will be addressed in section six.

II. Mathematical Model of the BLDC Motor

The motor drive circuit and the equivalent circuit of a three-phase delta-connected BLDC motor are depicted in Figure 1. The mathematical model of the motor is considered in ideal conditions. The voltages v_a, v_b, v_c across phases a, b, c can be expressed as (1) [8], [10],

$$\begin{aligned} v_a &= Ri_a + L\frac{di_a}{dt} + M\frac{di_b}{dt} + M\frac{di_c}{dt} + e_a \\ v_b &= Ri_b + M\frac{di_a}{dt} + L\frac{di_b}{dt} + M\frac{di_c}{dt} + e_b \\ v_c &= Ri_c + M\frac{di_a}{dt} + M\frac{di_b}{dt} + L\frac{di_c}{dt} + e_c \end{aligned} \quad (1)$$

where i_a, i_b, and i_c are three-phase currents; R is the phase resistance, L is the phase self-inductance, and M is the mutual inductance between phases. e_a, e_b, and e_c are the ideal three-phase BEMFs. These equations are independent and suitable for both wye and delta connections. In the delta connection, phase voltages are equal to line voltages.

Assume that the magnetic saturation is neglected. The self-inductance and mutual-inductance are supposed not be effected by current and rotor position. In a three-phase BLDC motor, the electromagnetic torque, T_e in N-m, can be expressed as (2) as the mechanical loss is ignored,

$$T_e = \frac{e_a i_a + e_b i_b + e_c i_c}{\omega_m} \quad (2)$$

where ω_m is the mechanical angular speed, in rad/s.

The electromechanical system is the relationship between the electromagnetic torque, the load torque, T_L in N-m, and the speed of the motor. It is known as the motion equation and can be presented by (3).

$$T_e = J\frac{d\omega_m}{dt} + B\omega_m + T_L \quad (3)$$

where J is the moment of inertia, in kg-m^2/sec^2; B is the viscous friction, in N-m/rad/sec.

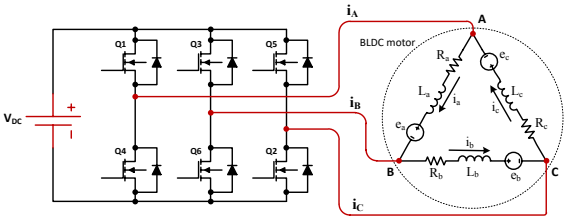


Figure 1. The equivalent circuit of a three-phase delta-connected BLDC motor drive.

III. The Sensorless Control Scheme

A. The Squarewave Drive with 120-Degree Conduction Mode

The control strategy is based on the 120^0 conduction interval. There are six 60^0 intervals entire an electrical cycle. Either the upper transistors are applied for chopping while the lower transistors are used to be switched on and off during its conducting interval or vice versa. Figure 2 illustrates the Hall-effect based position information, and the ideal waveforms of the delta-connected BLDC motor with its' the trapezoidal BEMF case [10]. It is noticed that the LL BEMF is also the same as the phase BEMF.

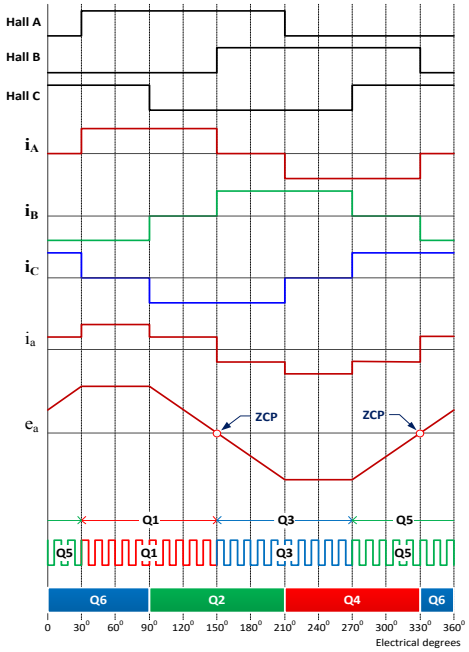


Figure 2. The ideal waveforms of the delta-connected BLDC motor.

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Figure 3 shows the current direction flow of the conduction interval from 30^0 to 90^0 in case the transistors Q1 and Q6 are conducted where the line current i_A and the phase current i_a are the same positive direction. The next state of the currents is controlled by the two transistors Q1 and Q2. Table I shows the summaries of commutation transistors and line currents of the BLDC motor [10].

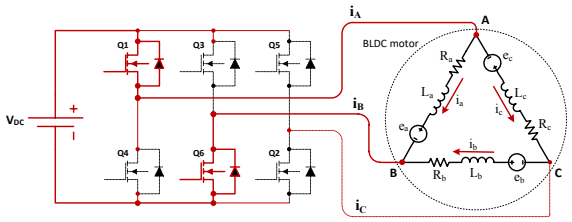


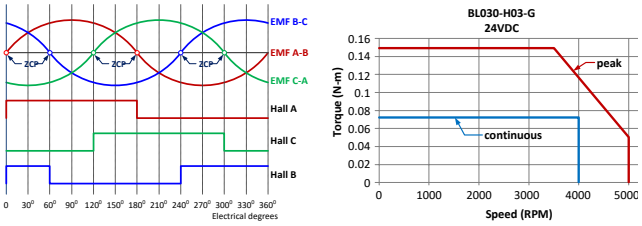
Figure 3. The current path of the conduction interval from 30^0 to 90^0.

TABLE I. SUMMARIES OF COMMUTATION TRANSISTORS AND LINE CURRENTS OF THE BLDC MOTOR

Rotor position	Commutation transistors						Line currents		
	Q1	*Q4*	*Q3*	*Q6*	*Q5*	*Q2*	i_A	i_B	i_C
$30^0 - 90^0$	1	0	0	1	0	0	+1	-1	0
$90^0 - 150^0$	1	0	0	0	0	1	+1	0	-1
$150^0 - 210^0$	0	0	1	0	0	1	0	+1	-1
$210^0 - 270^0$	0	1	1	0	0	0	-1	+1	0
$270^0 - 330^0$	0	1	0	0	1	0	-1	0	+1
$330^0 - 30^0$	0	0	0	1	1	0	0	-1	+1

B. *Design of Sensorless Circuit with Variable Cut-off Frequencies*

As mentioned in the Introduction section, it is impossible to measure the BEMF directly from the idle terminal voltage due to delta connection windings of the BLDC motor. Consequently, the sensorless circuit with Schmitt trigger circuit in [8] was implemented in this paper. Figure 4 shows the relation between phase EMFs and Hall sensor outputs at clockwise (CW), and the torque-speed curves of the delta-connected BLDC motor [11].



(a) (b)

Figure 4. (a) The relation between ideal phase BEMFs and the Hall output signals; (b) Torque-speed curves.

It can be seen that, the ZCPs of the line/phase EMF A-B are in phase with the Hall A output signal. Moreover, the output LL voltages of the inverter with 120^0 conduction are also in phase with the phase EMFs. Therefore, these features are used to implement of the sensorless circuit as shown in Figure 5 [8].

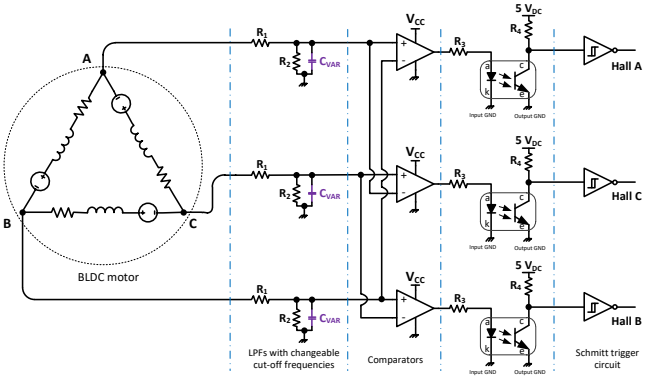


Figure 5. The SCSs detection circuit.

In general, for the sensorless scheme, a higher cut-off frequency is usually adopted to achieve a small phase delay angle at high-speed, but this may cause the poor quality of the SCSs at low speeds. To solve this problem, the small cut-off frequency is employed; however, this leads to a large phase delay as increasing in speed. Therefore, in order to overcome these problems, the phase delay or phase advance should be compensated by different algorithm approaches for specific applications. The cut-off frequency, f_c, and phase delay, φ_L, caused by the LPF can be calculated as (4), (5) [6]–[8].

$$f_c = \frac{1}{2\pi RC} \tag{4}$$

$$\varphi_L = \tan^{-1} CR2\pi\omega_e \tag{5}$$

where, $R = R_1R_2/(R_1 + R_2)$, R_1 and R_2 are the resistors of the LPF's voltage divider, C is the filter capacitance, ω_e is the BEMF frequency of the motor.

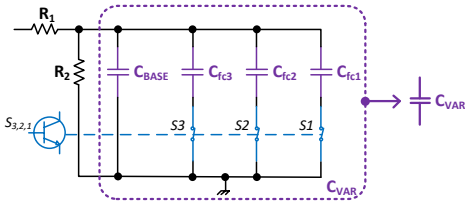


Figure 6. The proposed LPF with variable cut-off frequencies.

TABLE II. SUMMARIES OF THE CUT-OFF FREQUENCIES CORRESPONDING TO OPERATING SPEED

Operating Speed Range (RPM)	600-999	1000-1500	1501-2500	>2501
Cut-off frequency (Hz)	26.61	42.87	72.34	105.23
R_1 (kΩ)	22	22	22	22
R_2 (kΩ)	10	10	10	10
C (μF)	0.87	0.54	0.32	0.22

Accordingly, in this application, the analog LPF approach with the cut-off frequencies can be variable is designed as shown in Figure 6 to meet the torque-speed curves as depicted in Figure 4 (b). Each LPF has a bank of capacitors, including four capacitors, which are represented for four cut-off

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frequencies, connected in parallel. The C_{BASE} capacitor is designed to operate at the rated-speed 4000 rpm with the highest cut-off frequency. The remaining three capacitors are applied for the remainder cut-off frequencies in which the sum of values of the capacitors C_{BASE}, C_{fc3}, C_{fc2}, and C_{fc1} are used for the low operating speeds with the lowest cut-off frequency. The three capacitors, each of them in sequence, is wired to a transistor (act as a switch) in series in each its' parallel branch. Accordingly, the microcontroller sends the control signals to the transistors to adjust the cut-off frequency according to the motor speed. Table II shows a simple compensation algorithm and summaries of LPF's parameters and the four cut-off frequencies for a wide speed-range in this application.

IV. The System Architecture

The diagram of the hardware architecture is used in this paper as shown in Figure 7 [8]. It comprises a microcontroller 32-bit TM4C123GH6PM, which is integrated on a Tiva-C series Launchpad Evaluation Kit named is EK-TM4C123GXL from the Texas Instruments Inc., a motor drive circuit integrated of a three-phase inverter, buffer and isolation circuits, protection circuits, and sensorless circuit with variable cut-off frequencies.

The firmware in this application was written by C language based on the Integrated Development Environment (IDE) tool developed by Keil company. It includes a sensorless starting open-loop module, PWM module, speed and rotor position calculation modules, current measurement module. The speed commands were sent from a graphical user interface (GUI) by using the Microsoft Visual Studio 2017 C# development platform [8]. A digital PID controller with anti-windup was employed to control the speed of the motor. This is a speed closed-loop which would be switched to the sensorless mode as soon as the sensorless starting strategy was terminated.

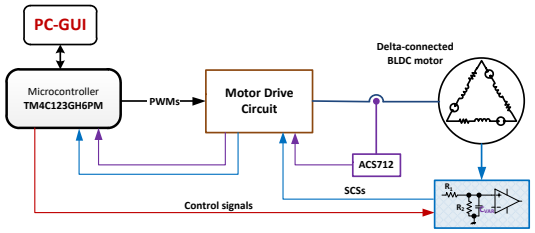


Figure 7. System hardware architecture diagram.

Regarding the algorithm of variable cut-off frequency implemented in this application, it is a simple algorithm as following. At starting, all capacitors were connected in parallel for increasing the capacitance of the capacitors by switching on all the transistors for easy starting purposes. This setting was applied for all speed commands of the operating speed range as shown in Table II. Then depending on the speed range command, the microcontroller would send control signals to keep the original starting condition if the speed range within the lowest speed region, or to switch transistors off corresponding to the requirement of speed increasing. It is noticed that, as speeding up, the transistors should be disconnected after the actual speed reached up the speed command. In contrast, as speeding down, the transistors should be switched on to obtain a low cut-off frequency appropriately before the speed command is executed.

V. Experimental Results and Discussions

The experimental testing platform of the delta-connected BLDC motor in this paper is illustrated in Figure 8. It contains a Tiva-C Launchpad EK-TM4C123GXL, a motor drive, two delta-connected BLDC motors coupled to each other in which one is a motor under test, the other acting as a DC generator load, and a DC power supply. The specifications of the delta-connected BLDC motor can be found at [11]. All the following results were tested under load to verify the feasibility and stability of the proposed system.

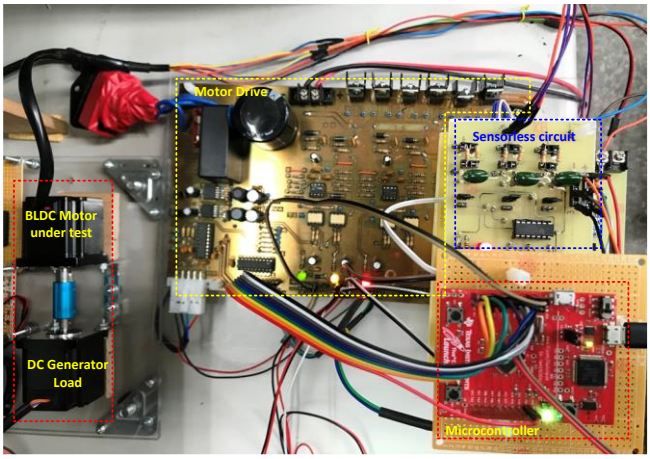


Figure 8. Experimental set up for the delta-connected BLDC.

Figure 9 shows the SCSs for phases A, B, and C and the ZCPs of the LL voltage between phases B and C. It can be seen that the SCSs were extracted from the proposed circuit in Figure 5 giving proper and stable commutation signals for the delta-connected BLDC motor. In addition, the SCS of phase A was compared to the Hall A sensor signal as shown in Figure 10. It indicates that the two commutation signals are almost identical and in phase to each other. These results verified that the proposed sensorless circuit was successfully used in this application.

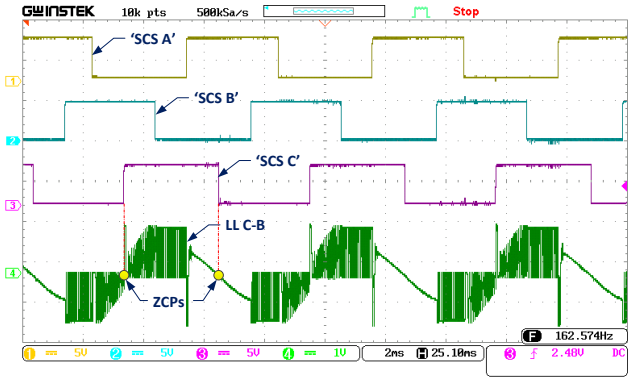


Figure 9. The waveforms of SCSs and ZCPs of LL between phases B and C.

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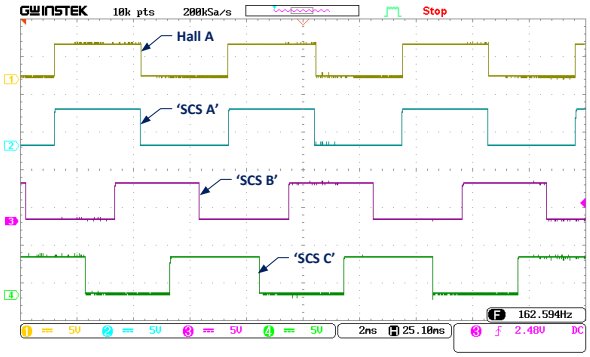


Figure 10. A comparison of commutation signals between Hall A sensor signal and SCS of the phase A.

In order to test the sensorless speed responses of the system as well as the performance of the PID algorithm, a speed step profile with a wide speed range for both stepping-up and stepping-down was performed as shown in Figure 11. It can be observed that the actual speed can catch up with the speed command in time.

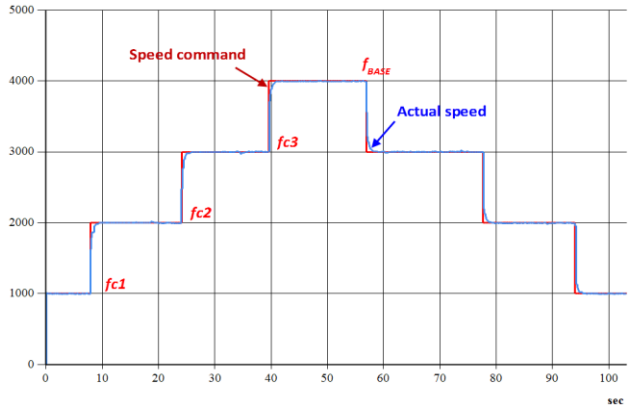
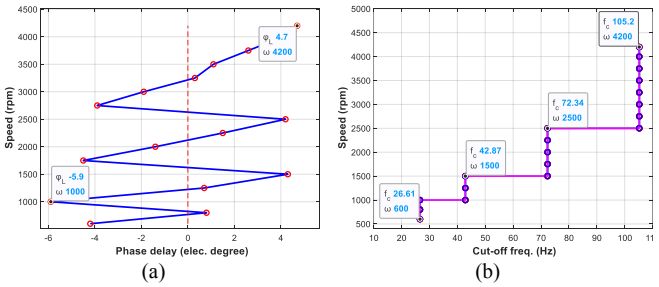


Figure 11. Speed step profile with a wide speed range for the sensorless mode.

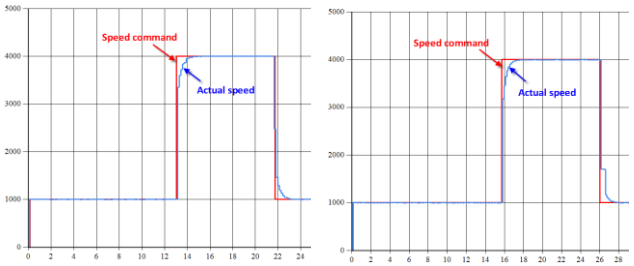
Figure 12 (a) shows the experimental results of phase delay angles with different operation cut-off frequencies for a wide speed range. It indicates that the phase delay caused by LPF controlled effectively to keep within acceptable commutation errors from -5.90 to +4.70. Figure 12 (b) illustrates the compensation algorithm corresponding to the relation between cut-off frequencies, and the speed ranges.



(a) (b)

Figure 12. Phase-delay behaviors and the compensation algorithm corresponding to a wide speed range for the sensorless mode.

Moreover, the speed step response with a large step for both stepping-up and stepping-down, from 1000 rpm to 4000 rpm and vise-versa, was performed to verify the quality of the PID controller as well as the feasibility of the sensorless application as shown in Figure 13. Figure 13 (a) represents the speed step responses from the Hall sensor mode, while Figure 13 (b) illustrates for sensorless mode. It can be seen that both modes give the similar good performance, the time for the actual speed reaches up the speed command is less than 1 second.



(a) With (b) Without

Figure 13. A large speed step response with and without Hall-effect sensors.

Finally, to validate and compare the performance of the two control modes with and without Hall effect sensors for the delta-connected BLDC motor, current values of the line A were measured for the whole speed range for different cut-off frequencies of the proposed LPF as presented in Figure 14. Accordingly, it can be found that at speeds around 2500 rpm then the currents from the sensorless control mode are slightly less than that of the Hall control mode. However, in general, the performance of the sensorless mode is met with that of the Hall control mode in this application, especially in low-speed regions.

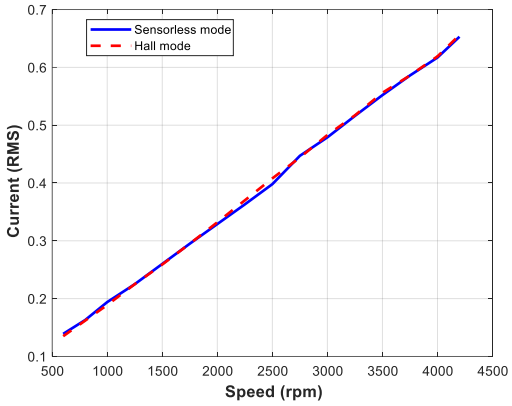


Figure 14. Comparison of the line currents between the Hall and sensorless modes.

VI. Conclusions

In this paper, a sensorless motor drive for a delta-connected BLDC motor rather than a wye-connected BLDC motor was designed and fulfilled successfully. The sensorless circuit was based on the line-to-line terminal voltages implemented effectively instead of extracting the phase

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BEMF from the idle phase. Moreover, the sensorless compensation algorithm used is simple by changing the cut-off frequencies of the LPFs according to the changes in speeds. To alternate the cut-off frequency, the transistors were employed to connect or disconnect the capacitors in parallel for varying capacitor capacitance and thus changing in the cut-off frequency. Furthermore, in order to evaluate the performance of the proposed sensorless control, the values of line A current was measured to compare with that of the Hall sensor-based control. Finally, the experimental results proved the feasibility and validation of the proposed sensorless method in this paper. The results showed that the proposed control achieved good performance in this application.

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